Tumour-Core, Peritumoral-Margin, and Remote-Context Evidence in Brain MRI Classification: A Sham-Controlled, Plane-Stratified Counterfactual Audit Across CNN and Vision Transformer Models

Running Title: Region-Resolved Counterfactual Evidence in Brain Tumour MRI
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Structured Abstract

Background

Deep convolutional neural networks (CNNs) and Vision Transformers (ViTs) regularly report high benchmark accuracy for classifying primary intracranial neoplasms from contrast-enhanced magnetic resonance imaging (CE-MRI). However, standard evaluation protocols evaluate models exclusively on unmodified images, failing to disentangle whether high diagnostic accuracy reflects genuine reliance on intra-lesional tumour morphology, opportunistic exploitation of peritumoral transitional zones, or spurious shortcut learning from remote cranial background context and scanner acquisition artifacts (the "Clever Hans" effect).

Methods

We establish a literature-informed, fully reproducible counterfactual evaluation framework to audit region-resolved diagnostic evidence across the Cheng brain tumour MRI cohort (N = 3,064 slices, 233 patients) and the multi-plane BRISC benchmark (N = 6,000 slices). Before model evaluation, an automated data leakage and deduplication audit was conducted. In Cheng, a 5-fold cross-validation scheme was strictly grouped by patient ID, guaranteeing zero patient overlap across folds; baseline validation was evaluated on the patient-isolated held-out Fold 1 test partition (N = 480 slices, 49 patients). In BRISC, an automated perceptual hashing and SHA-256 pixel matrix audit revealed 159 exact duplicate image pairs; among these, 110 pairs cross the official train-test split boundary, involving 101 unique test slices (10.1% of the official 1,000-slice test set; 96 meningiomas, 5 pituitary adenomas). In light of published deduplication claims in the BRISC data descriptor, we treat this finding as an archive-specific empirical observation and report both the official benchmark (N = 1,000) and a de-duplicated clean benchmark (N = 899; 759 mask-supported), alongside complete prediction exports. We trained ImageNet-pretrained ResNet-50 (CNN) and DeiT-Small/16 (ViT) architectures using AdamW (batch size 16, weight decay 10^{-2}, initial learning rate 10^{-4} for ResNet-50 with dropout 0.2, 5 × 10^{-5} for DeiT-Small with dropout 0.1, 1-epoch linear warmup, early stopping patience 3, max 5 epochs). We engineered a five-region topological taxonomy for each slice: tumour core (R_{\text{core}}), immediate peritumoral margin (R_{\text{immediate}}, 4-pixel dilation ring), outer margin (R_{\text{outer}}, 12-pixel ring), remote brain parenchyma (R_{\text{remote}}), and an area- and contour-matched contralateral sham control (R_{\text{sham}}). Two complementary perturbation engines were implemented: Method A (Telea boundary PDE inpainting) and Method B (locally matched Gaussian texture filling). In total, 1,200 counterfactual interventions were verified with 100% strict non-target pixel invariance (|Δ| < 10^{-5} outside mask), covering all 90 evaluated slices. We evaluated 1,800 paired counterfactual observations across models and datasets, computing raw sensitivity ($\Delta_r = p_y(x) - p_y(x_{\text{edit}, r})$) and sham-corrected region sensitivity ($\tau_r = \Delta_r - \Delta_{\text{sham}}$), paired Wilcoxon signed-rank tests with Holm-Bonferroni correction, patient-clustered analyses across patient clusters, and Cohen's d_z effect sizes.

Results

Perturbing the tumour core produced the largest evidence drop under boundary PDE inpainting across both models ($p < 0.05$). On Cheng ($N = 45$ slices across $N = 10$ patient clusters), DeiT-Small demonstrated a sham-corrected sensitivity of $\tau_{\text{core}} = 0.1282 \pm 0.248$ (slice-level Wilcoxon $p = 9.13 \times 10^{-6}$, Holm-adjusted $p = 2.74 \times 10^{-5}$, patient-clustered Wilcoxon $p = 0.0488$, $d_z = 0.518$) under inpainting and $\tau_{\text{core}} = 0.1449 \pm 0.217$ under texture filling. ResNet-50 on Cheng showed $\tau_{\text{core}} = 0.0311 \pm 0.057$ (slice-level $p = 6.54 \times 10^{-4}$, Holm-adjusted $p = 1.96 \times 10^{-3}$, patient-clustered $p = 0.0488$, $d_z = 0.548$) under inpainting. On BRISC ($N = 45$), DeiT-Small demonstrated $\tau_{\text{core}} = 0.0517 \pm 0.120$ (raw drop 0.0367 ± 0.096, Wilcoxon $p = 0.0320$, $d_z = 0.432$) under inpainting. In a pooled multi-cohort comparison ($N = 90$, Method A), DeiT-Small exhibited significantly greater core evidence reliance than ResNet-50 ($\tau_{\text{core, ViT}} = 0.0900$ vs $\tau_{\text{core, CNN}} = 0.0161$, difference $+0.0738$, slice-level Wilcoxon $p = 0.0123$, cluster-level Wilcoxon $p = 0.0162$ across 55 independent patient/case clusters, $d_z = 0.362$) alongside superior probability calibration (Cheng 10-bin Expected Calibration Error: DeiT-Small 0.0366 vs ResNet-50 0.1835). Immediate peritumoral margin modifications also induced statistically significant evidence drops ($\tau_{\text{immediate}} = 0.0213 \pm 0.096$, $p = 0.0267$ on Cheng; $\tau_{\text{immediate}} = 0.0239 \pm 0.063$, raw drop 0.0089 ± 0.048, $p = 0.0152$, Holm-adjusted $p = 0.0455$ on BRISC for DeiT-Small). Spatial evidence decay was condition- and method-dependent: while core dominance held strongly under inpainting, outer margin (12-pixel ring) edits under Gaussian texture filling and in skull-stripped BRISC scans produced effects comparable to the core (BRISC DeiT outer margin $\tau = 0.0301 \pm 0.071$, $p = 0.0049$), reflecting both large annular mask tissue volume covering vasogenic edema and convolutional sensitivity to high-frequency boundary textures. Remote parenchyma edits yielded negligible shifts indistinguishable from sham controls ($\tau_{\text{remote}} = 0.0140$, $p = 0.2815$ on BRISC; $\tau_{\text{remote}} = 0.0128$, $p = 0.0801$ on Cheng). Exploratory radiologic analysis showed that gliomas exhibited a substantial margin-to-core sensitivity ratio (35.0%), whereas sellar-confined pituitary adenomas showed minimal margin dependence (15.1%). Finally, duplicate test images in BRISC shifted test class priors, altering benchmark macro-F1 (Full Clean Test: DeiT-Small 0.8344 vs ResNet-50 0.6991; Neoplasm Clean Batch: DeiT-Small 0.5999 vs ResNet-50 0.4181).

Discussion

Contrast-enhanced brain tumour classifiers derive diagnostic evidence primarily from intra-tumoural morphology, with non-trivial contributions from immediate transitional margins, while remaining robust to remote context. ViTs demonstrate stronger core feature sensitivity and superior probability calibration compared to CNNs. However, duplicate leakage and class-prior shifts in public benchmarks underscore the critical necessity of rigorous data auditing, patient-clustered evaluation, and transparent sample accounting before clinical interpretation.

1. Introduction

Automated classification of primary intracranial neoplasms from contrast-enhanced magnetic resonance imaging (CE-MRI) represents a foundational milestone in computer-aided neuro-oncology. Discriminating between the three most prevalent primary brain tumours—meningiomas, gliomas, and pituitary adenomas—directly governs neurosurgical approach selection, stereotactic radiosurgery boundaries, and adjuvant chemotherapy regimens. Over the past decade, deep learning systems based on convolutional neural networks (CNNs) and Vision Transformers (ViTs) have repeatedly reported benchmark classification accuracies exceeding 90% to 95% on public MRI datasets [2, 5].

Despite these impressive nominal scores, translational deployment of medical imaging AI into clinical radiology workflows remains severely impeded by vulnerability to shortcut learning and the "Clever Hans" effect [1, 11]. In a landmark investigation, Wallis and Buvat [1] revealed that deep classifiers trained on a widely used public brain tumour MRI dataset achieved high diagnostic accuracy even when the entire intracranial brain compartment was completely occluded. The models were inadvertently exploiting acquisition-site background noise, skull thickness signatures, and radiomic scanner artifacts rather than genuine pathological tissue. Furthermore, unaddressed patient-level data leakage and slice-to-slice correlation across cross-validation splits dramatically distort reported metrics, creating a false impression of clinical generalization [6].

Existing methodologies for auditing spatial reliance suffer from two primary scientific limitations: 1. **Coarse Binary Masking & Out-of-Distribution Shift:** Prior occlusion studies predominantly replace entire cerebral hemispheres or rectangular tiles with zero-intensity black patches or uniform noise [3, 4]. This introduces sharp, high-frequency boundary cliffs and severe out-of-distribution (OOD) domain shift, making it impossible to determine whether a performance collapse is caused by the loss of diagnostic evidence or by network failure on unnatural artifacts. 2. **Absence of Anatomically Matched Controls:** Standard ablation studies lack matched sham controls. Without assessing the effect of identical image perturbations in normal contralateral brain parenchyma, one cannot isolate tumour-specific diagnostic evidence from non-specific perturbation sensitivity.

Research Objectives and Hypotheses

To address these challenges, we design and execute a literature-informed, sham-controlled, plane-stratified counterfactual audit across CNN (ResNet-50) and Vision Transformer (DeiT-Small/16) architectures (Figure 1). We investigate five central scientific hypotheses: - **Hypothesis 1 (Spatial Evidence Gradient):** Model prediction sensitivity is concentrated in intra-lesional core and transitional margin regions, whereas remote tissue perturbations produce negligible shifts indistinguishable from sham controls ($p > 0.05$). - **Hypothesis 2 (Transitional Margin Contribution):** Immediate peritumoral margin modifications (4-pixel annular ring) produce statistically significant evidence drops ($p < 0.05$), reflecting clinical peripheral contrast enhancement and infiltrative border morphology. - **Hypothesis 3 (Architectural Inductive Bias):** Localized convolutional kernels (ResNet-50) and global self-attention mechanisms (DeiT-Small) exhibit divergent spatial reliance profiles, with ViTs demonstrating greater intra-lesion core reliance and superior probability calibration. - **Hypothesis 4 (Perturbation Method Concordance):** Regional sensitivity rankings are concordant across distinct

inpainting mechanisms (boundary PDE vs Gaussian texture fill). - **Hypothesis 5 (Benchmark Leakage & Prior Distortion):** Public multi-center benchmarks contain duplicate leakage across train and test partitions that distorts class priors and standard evaluation metrics.

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2. Related Work

2.1 Shortcut Learning in Medical Image Analysis

Shortcut learning occurs when machine learning models achieve high nominal accuracy by exploiting spurious decision rules that fail under distributional shift [11]. In radiographic COVID-19 detection, DeGrave et al. [14] demonstrated that deep networks relied heavily on hospital-specific radiopaque markers and scanner-bed textures rather than pulmonary consolidations. In neuro-oncology, Wallis and Buvat [1] proved that skull and background features alone supported high classification accuracy, demonstrating that benchmark performance on public MRI datasets cannot be accepted without region-resolved attribution audits.

2.2 Spatial Interventions and Counterfactual Attribution

Attribution by input perturbation has evolved from simple occlusion sensitivity to generative counterfactuals. Goyal et al. [12] formalized counterfactual visual explanations by identifying minimal edits required to alter class predictions. Rossi et al. [3] introduced TACE (Tumour-Aware Counterfactual Explanations) as a preprint framework to evaluate classifier behavior under generative lesion insertion and removal. Sourget et al. [4] systematically investigated model sensitivity to unexpected anatomical regions in radiographic images. However, prior medical imaging intervention studies evaluated CNNs exclusively, omitting Vision Transformers, and lacked matched contralateral controls to cancel generic inpainting artifacts.

2.3 Brain-Tumour MRI Classification & Benchmark Leakage Audits

The brain tumour dataset published by Cheng et al. [2] (N = 3,064 slices) has served as a foundational benchmark for over a decade. Recently, Fateh et al. [5] introduced the BRISC benchmark (N = 6,000 multi-plane slices). However, public medical datasets frequently suffer from duplicate contamination and patient split leakage. As demonstrated by Yagis et al. [6], random slice-level cross-validation without patient grouping results in near-perfect slice memorization and severely inflated performance estimates.

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3. Datasets, Provenance & Data Leakage Audit

3.1 Datasets and Cohort Overview

We evaluated two contrast-enhanced T1-weighted brain MRI datasets: 1. **The Cheng Dataset [2]:** 3,064 CE-T1 slices from 233 unique patients acquired across three tumour classes: 708 meningiomas, 1,426 gliomas, and 930 pituitary adenomas. Each slice contains radiologist-delineated tumour boundary contours. 2. **The BRISC Dataset [5]:** 6,000 skull-stripped CE-MRI slices partitioned into an official training set (N = 5,000, split into 4,000 train and 1,000 validation) and an official test set (N = 1,000)

across 4 classes: glioma, meningioma, pituitary tumour, and no-tumour controls, imaged in Axial, Coronal, and Sagittal planes. Usable segmentation masks are available for 860 test neoplasm slices.

3.2 Sample Accounting and Selection Rules

To ensure complete transparency and reproducibility across all reported tables and figures, the exact sample flow and selection rules are structured as follows: - **Baseline Evaluation Partitions:** - For Cheng (N = 3,064 slices across 233 patients), a 5-fold patient-isolated cross-validation split was created. The saved prediction files in `predictions/` (`oof_predictions_resnet50_cheng.csv` and `oof_predictions_deit_small_patch16_224_cheng.csv`) contain exactly N = 480 **records, all belonging to Fold 1** (evaluated across 49 unseen patients). Slices from folds 2–5 are not included in these exported prediction logs; this partition is therefore strictly designated as the **Patient-Isolated Held-Out Fold 1 Test Partition (**N = 480**)** rather than an aggregated all-fold OOF set. - For BRISC, the complete official test predictions (N = 1,000) are exported in `predictions/test_predictions_resnet50_brisc.csv` and `predictions/test_predictions_deit_small_patch16_224_brisc.csv`, while the clean de-duplicated predictions (N = 899) are exported in `predictions/test_predictions_resnet50_brisc_clean.csv` and `predictions/test_predictions_deit_small_patch16_224_brisc_clean.csv`. The standardized mask-supported neoplasm evaluation subset (N = 480; 254 gliomas, 226 meningiomas) is preserved in `predictions/test_predictions_*_neoplasm_480.csv`. - **Counterfactual Intervention QC Manifest (**N = 1,200**):** Recorded in `manifests/intervention_qc.csv` across 120 unique slices (60 Cheng, 60 BRISC) across 5 regions (R_{\text{core}}, R_{\text{immediate}}, R_{\text{outer}}, R_{\text{remote}}, R_{\text{sham}}) using 2 perturbation engines (Method A: Telea inpaint; Method B: Gaussian texture fill), yielding 120 × 5 × 2 = 1,200 verified records. Crucially, **100% of the 90 evaluated slices (45 Cheng + 45 BRISC) have complete, verified QC records in this manifest**, confirming zero non-target corruption (|Δ| < 10^{-5}). - **Paired Counterfactual Observations (**N = 1,800**):** Evaluated across 45 matched patient slices per dataset (N = 90 total) across 5 regions, 2 perturbation methods, and 2 model architectures (ResNet-50 and DeiT-Small). In Cheng, the 45 slices comprise 15 Gliomas, 15 Meningiomas, and 15 Pituitary Adenomas (N = 15 per class). Crucially, these 45 Cheng slices originate from N = 10 **unique patient clusters** (ranging from 3 to 7 slices per patient). In BRISC, the 45 slices comprise 15 Axial, 15 Coronal, and 15 Sagittal slices (N = 15 per acquisition plane) across the three neoplasm classes. Total evaluated paired rows = 90 \text{ slices} × 5 \text{ regions} × 2 \text{ methods} × 2 \text{ architectures} = 1,800 rows in `predictions/intervention_effects.csv`.

3.3 Duplicate Audit and Split Policy

- **Cheng Patient-Isolated 5-Fold Cross-Validation:** Embedded MATLAB struct headers were parsed to extract patient identifiers (`PID`). A 5-fold cross-validation split was established using `StratifiedGroupKFold(n_splits=5, shuffle=True, random_state=42)` on patient ID. Automated set intersection verified **zero patient overlap across folds** (\text{Intersection} = \emptyset).
- **BRISC Duplicate Discovery and Provenance:** In our local evaluated archive files (`manifests/duplicate_pairs.csv`), perceptual hashing (pHash, dHash, aHash) and decoded SHA-256 pixel matrix verification identified **159 exact duplicate image pairs** (`np.array_equal == True`). Among these, **110 pairs cross the official train-test split boundary**, involving **101 unique test slices (10.1% of the 1,000-slice test set)** that are byte-identical copies of training/validation

images (96 meningiomas and 5 pituitary adenomas). However, the peer-reviewed data descriptor by Fateh et al. [5] (*Scientific Data*, 2026) reports that deduplication was conducted prior to dataset partitioning, and other external replication audits report varying hash-cleaning counts. We therefore present this local finding as an empirical observation specific to the downloaded archive release rather than a definitive indictment of the published benchmark design. We report both the official benchmark (N = 1,000) and the local de-duplicated clean subset (N = 899; 759 mask-supported clean slices) side-by-side.

- **Cross-Dataset Overlap:** Deduplication screening between Cheng (3,064 slices) and BRISC (6,000 slices) confirmed **0 duplicate pairs** and structural similarity index (SSIM) values bounded between 0.28 and 0.58, confirming independent clinical cohorts.

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4. Materials and Methods

4.1 Image Preprocessing and Geometric Standardization

All slices were standardized through an audited pipeline (Figure 2): 1. **Percentile Intensity Clamping:** Tissue intensities were mapped to [0.0, 1.0]:

$$\tilde{x} = \text{clip}\left(\frac{x - P_1(x)}{P_{99}(x) - P_1(x)}, 0.0, 1.0\right) \tag{1}$$

where P_1 and P_{99} denote the 1st and 99th percentiles of intracranial tissue. 2. **Aspect-Ratio-Preserving Letterbox Padding:** Slices were scaled along their maximum axis to 224 pixels and symmetrically padded with zeros to 224 × 224. Nearest-neighbor interpolation was enforced for segmentation masks. Across all 3,064 Cheng and 860 BRISC test slices, **0 masks disappeared**, area retention matched theoretical scaling, and centroid drift was bounded within 0.0035. 3. **Channel Standardization:** Slices were replicated to 3 channels and normalized using ImageNet parameters ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). Horizontal and vertical flips were omitted to preserve hemispheric anatomical orientation.

4.2 Model Architectures and Training Protocols

Models were configured in strict concordance with `configs/training.yaml`: - **ResNet-50 (CNN):** 23.51 million parameters; initialized with `IMAGENET1K_V2` weights [7]; linear classification head; Dropout p=0.2. Initial learning rate 1×10^{-4}. - **DeiT-Small/16 (ViT):** 21.66 million parameters; initialized with ImageNet-1k weights [8]; 12 Transformer blocks, 384 embedding dimension, 6 heads; class token projection head; Dropout p=0.1. Initial learning rate 5×10^{-5}. - **Optimization:** Both architectures were optimized using AdamW with weight decay 1×10^{-2} and batch size 16. Training ran for up to 5 epochs per fold/split with early stopping patience of 3 epochs on validation macro-F1, following a 1-epoch linear warmup and cosine annealing schedule. Standard cross-entropy loss was minimized with random seed 42.

4.3 Five-Region Topological Decomposition

For each test slice x with tumour mask M and intracranial tissue support S: 1. **Tumour Core (R_{core}):** The lesion mask M. 2. **Immediate Peritumoral Margin ($R_{\text{immediate}}$):** 4-pixel dilation ring:

$$R_{\text{immediate}} = (M \oplus K_4 \setminus M) \cap S \tag{2}$$

3. **Outer Peritumoral Margin (R_{outer}):** 12-pixel dilation ring:

$$R_{\text{outer}} = (M \oplus K_{16} \setminus (M \oplus K_4)) \cap S \tag{3}$$

4. **Remote Context (R_{remote}):** Tissue patch located > 32 pixels from the tumour boundary within S.

5. **Matched Contralateral Sham Control (R_{sham}):** Area- and contour-matched control placed in contralateral normal parenchyma:

$$R_{sham} \cap (R_{core} \cup R_{\text{immediate}} \cup R_{\text{outer}}) = \emptyset \tag{4}$$

4.4 Dual Counterfactual Perturbation Engines

- **Method A (Telea PDE Inpainting):** Solves the Fast Marching Method boundary PDE (`cv2.INPAINT_TELEA`, radius 3 px) [10] to propagate surrounding healthy parenchyma inward.
- **Method B (Locally Matched Gaussian Texture Fill):** Estimates local surrounding mean μ_{local} *and variance* σ^2_{local} from a 5-pixel dilated border, filling the region with $\mathcal{N}(\mu_{\text{local}}, \sigma_{\text{local}})$ smoothed via a 2-pixel cosine border taper.
- **Non-Target Pixel Invariance:** Both methods strictly guarantee bitwise preservation:

$$x_{\text{edit}}(u, v) \equiv x(u, v) \quad \forall\ (u, v) \notin R \tag{5}$$

4.5 Sensitivity Estimands and Statistical Inference

- **Raw Evidence Drop (Δ_r):** $\Delta_r(x) = p_y(x) - p_y(x_{\text{edit}, r})$.
- **Sham-Corrected Region Sensitivity (τ_r):** $\tau_r(x) = \Delta_r(x) - \Delta_{\text{sham}}(x)$.
- **Top-1 Prediction Flip (F_r):** $F_r(x) = \mathbb{I}[\arg\max p(x_{\text{edit}, r}) \neq \arg\max p(x)]$.
- **Hypothesis Testing:** Two-sided paired Wilcoxon signed-rank tests testing H_0: $\Delta_r = \Delta_{\text{sham}}$ with Holm-Bonferroni family-wise multiplicity adjustment across evaluated regions.
- **Patient-Clustered Analysis:** For the Cheng cohort where 45 slices originate from 10 patients, patient-level mean sensitivities were computed and evaluated with clustered Wilcoxon and clustered one-sample t-tests. For the pooled multi-cohort ($N = 90$), clustered inference across 55 independent clusters (10 Cheng patients + 45 BRISC cases) was conducted alongside Linear Mixed-Effects Modeling.
- **Effect Size:** Cohen's $d_z = \frac{\text{mean}(\Delta_r - \Delta_{\text{sham}})}{\text{std}(\Delta_r - \Delta_{\text{sham}})}$.

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5. Experimental Results

5.1 Baseline Classification Performance and Probability Calibration

Baseline models demonstrated strong discrimination across both benchmarks (Table 1; Figure 3). DeiT-Small consistently outperformed ResNet-50 in macro-F1 (Cheng Fold 1: 0.8403 vs 0.7190; BRISC Clean Full Test: 0.8344 vs 0.6991) and probability calibration. On Cheng Fold 1, DeiT-Small achieved a 10-bin Expected Calibration Error (ECE) of 0.0366 compared to 0.1835 for ResNet-50.

Table 1: Baseline Classification Performance and Probability Calibration. | Dataset | Model Architecture | Evaluation Partition | Accuracy | Balanced Acc | Macro-F1 | ECE (10-bin) | Brier Score | |---|---|---|---|---|---|---|---| | **Cheng** | ResNet-50 (CNN) | Patient-Isolated Held-Out Fold 1 (N = 480) | 0.7583 | 0.7448 | 0.7190 | 0.1835 | 0.3692 | | **Cheng** | DeiT-Small (ViT) | Patient-Isolated Held-Out Fold 1 (N = 480) | **0.8458** | **0.8394** | **0.8403** | **0.0366** | **0.2136** | | **BRISC** | ResNet-50 (CNN) | Official Test (N = 1,000, Full 4-Class) | 0.7040 | 0.6870 | 0.7095 | 0.3875 | 0.6510 | | **BRISC** | ResNet-50 (CNN) | Clean Test (N = 899, De-duplicated) | 0.6963 | 0.6815 | 0.6991 | 0.3790 | 0.6505 | | **BRISC** | DeiT-Small (ViT) | Official Test (N = 1,000, Full 4-Class) | 0.7880 | 0.8094 | 0.8030 | 0.0719 | 0.2864 | | **BRISC** | DeiT-Small (ViT) | Clean Test (N = 899, De-duplicated) | **0.8265** | **0.8330** | **0.8344** | **0.0901** | **0.2448** | | *BRISC (Sub)* | ResNet-50 (CNN) | Neoplasm Batch (N = 480; 2-Class) | 0.6479 | 0.6510 | 0.3460 | 0.3312 | 0.6591 | | *BRISC (Sub)* | ResNet-50 (CNN) | Clean Neoplasm Batch (N = 397) | 0.6247 | 0.6349 | 0.4383 | 0.3061 | 0.6595 | | *BRISC (Sub)* | DeiT-Small (ViT) | Neoplasm Batch (N = 480; 2-Class) | 0.6875 | 0.6813 | 0.3566 | 0.0421 | 0.3938 | | *BRISC (Sub)* | DeiT-Small (ViT) | Clean Neoplasm Batch (N = 397) | 0.7456 | 0.7294 | 0.3716 | 0.0718 | 0.3303 |

*Note: For Cheng, evaluations reflect the 480 held-out slices of Fold 1 (from 49 patients unseen in training). For BRISC, official and clean benchmark scores are computed across the full 4-class test partitions (*N = 1,000 *and* N = 899*, preserved in* `test_predictions_*_brisc.csv` *and* `test_predictions_*_brisc_clean.csv`*), alongside the mask-supported neoplasm evaluation subset (*N = 480 / N = 397*, preserved in* `test_predictions_*_neoplasm_480.csv`*).*

5.2 Quality Control of Counterfactual Interventions

All 1,200 generated counterfactual interventions achieved 100% compliance with quality-control gates (Figure 10): - **Non-Target Preservation:** Maximum absolute difference outside the edit mask was 0.00 × 10^0 across all 1,200 edits. - **Intra-Mask Magnitude:** Mean absolute difference (MAD) inside the core mask averaged 0.156–0.232 (Method A) and 0.216–0.259 (Method B). - **Evaluated Coverage:** 100% of all 90 evaluated slices in `predictions/intervention_effects.csv` possess verified QC records in `manifests/intervention_qc.csv`. - **Edge Discontinuity:** Boundary gradient offsets were bounded (0.16–0.24), confirming artifact-free transitions.

5.3 Primary Counterfactual Sensitivity: Core vs Margin Evidence

Evaluation of 1,800 paired counterfactual observations in `predictions/intervention_effects.csv` revealed key spatial sensitivities across architectures and intervention methods (Table 2; Figure 4).

Table 2: Regional Evidence Sensitivity and Biostatistical Hypothesis Testing (Recalculated from Canonical Raw Data). | Dataset | Model | Method | Region | Raw Sensitivity (Δ_r) | Sham-Corrected (τ_r) | Flip Rate | Raw Wilcoxon p | Holm-Adjusted p | Cohen's d_z | |---|---|---|---|---|---|---|---|---|---|---| | **Cheng (**N = 45**)** | DeiT-Small | Method A (Telea) | **Core** | 0.1212 ± 0.244 | 0.1282 ± 0.248 | 13.3% | 9.13 × 10^{-6} | 2.74 × 10^{-5} | **0.518** | | **Cheng (**N = 45**)** | DeiT-Small | Method B (Texture) | **Core** | 0.1577 ± 0.241 | 0.1449 ± 0.217 | 8.9% | 3.09 × 10^{-8} | 1.24 × 10^{-7} | **0.669** | | **Cheng (**N = 45**)** | DeiT-Small | Method A (Telea) | **Immediate (4px)** | 0.0143 ± 0.082 | 0.0213 ± 0.096 | 8.9% | **0.0267** | 0.0535 | **0.222** | | **Cheng (**N = 45**)** | DeiT-Small | Method A (Telea) | Outer (12px) | -0.0007 ± 0.091 | 0.0063 ± 0.098 | 8.9% | 0.2866 | 0.2866 | 0.064 (n.s.) | | **Cheng (**N = 45**)** | DeiT-Small | Method A (Telea) | Remote Context | 0.0058 ± 0.030 | 0.0128 ± 0.065 | 2.2% | 0.0801 | 0.1602 | 0.196 (n.s.) | | **Cheng (**N = 45**)** | ResNet-50 | Method A (Telea) | **Core** | 0.0264 ± 0.041 | 0.0311 ± 0.057 | 8.9% | 6.54 × 10^{-4} | 1.96 × 10^{-3} | **0.548** | | **Cheng (**N = 45**)** | ResNet-50 | Method B (Texture) | **Core** | 0.0844 ± 0.134 | 0.0603 ± 0.152 | 31.1% | **0.0173** | 0.0519 | **0.396** | | **Cheng (**N = 45**)** | ResNet-50 | Method A (Telea) | Immediate (4px) | 0.0079 ± 0.036 | 0.0125 ± 0.050 | 6.7% | 0.0621 | 0.1242 | 0.248 | | **BRISC (**N = 45**)** | DeiT-Small | Method A (Telea) | **Core** | 0.0367 ± 0.096 | 0.0517 ± 0.120 | 8.9% | **0.0320** | 0.0639 | **0.432** | | **BRISC (**N = 45**)** | DeiT-Small | Method A (Telea) | **Immediate (4px)** | 0.0089 ± 0.048 | 0.0239 ± 0.063 | 6.7% | **0.0152** | **0.0455** | **0.381** | | **BRISC (**N = 45**)** | DeiT-Small | Method A (Telea) | **Outer (12px)** | 0.0150 ± 0.065 | 0.0301 ± 0.071 | 11.1% | **0.0049** | **0.0197** | **0.425** | | **BRISC (**N = 45**)** | DeiT-Small | Method A (Telea) | Remote Context | -0.0010 ± 0.018 | 0.0140 ± 0.056 | 6.7% | 0.2815 | 0.2815 | 0.249 (n.s.) | | **Pooled (**N = 90**)** | DeiT-Small | Method A (Telea) | **Core** | 0.0789 ± 0.189 | 0.0900 ± 0.197 | 11.1% | 7.45 × 10^{-6} | 2.98 × 10^{-5} | **0.456** | | **Pooled (**N = 90**)** | ResNet-50 | Method A (Telea) | **Core** | 0.0129 ± 0.034 | 0.0161 ± 0.046 | 12.2% | 2.63 × 10^{-3} | 5.26 × 10^{-3} | **0.352** |

Non-Monotonic Spatial Dynamics and Patient-Clustering Analysis: 1. **Core Evidence Dominance vs Method Dependency:** Core dominance is strongly evident in Vision Transformers under PDE inpainting on Cheng, where core sensitivity (\tau_{\text{core}} = 0.1282) far exceeds margin effects (\tau_{\text{immediate}} = 0.0213, \tau_{\text{outer}} = 0.0063, \tau_{\text{remote}} = 0.0128). However, spatial decay is not universally monotonic across all settings: under Method B (locally matched Gaussian texture fill) and in skull-stripped BRISC scans, outer margin effects frequently equal or approach core sensitivity (e.g. BRISC DeiT outer \tau = 0.0301 vs core \tau = 0.0517). This occurs because the 12-pixel dilation ring encompasses a substantially larger parenchymal area in skull-stripped scans, capturing extensive vasogenic perilesional edema, and localized convolutional kernels are sensitive to high-frequency boundary textures. 2. **Patient-Clustered Statistical Inference:** In Cheng, the 45 intervention slices originate from N = 10 unique patient clusters. While slice-level Wilcoxon tests yield p = 9.13 × 10^{-6} for DeiT-Small and p = 6.54 × 10^{-4} for ResNet-50, patient-clustered analysis across the 10 patient-mean sensitivities yields clustered Wilcoxon p = 0.0488 for both architectures (clustered one-sample t-test: DeiT p = 0.0534, ResNet p = 0.0771). Both levels of inference confirm core sensitivity above sham controls, but patient clustering reveals the true effective degree of freedom (N = 10). 3. **Remote Context Invariance:** Remote parenchyma interventions produced small shifts indistinguishable from

sham controls ($p > 0.05$). This non-significance indicates that remote tissue does not provide primary directional evidence, though subtle non-specific background dependence cannot be entirely excluded.

5.4 Architectural Inductive Bias: ViT vs CNN

Paired comparison between DeiT-Small and ResNet-50 demonstrated key architectural divergences (Figure 5): - **Intra-Lesion Core Reliance:** In the pooled multi-cohort comparison (N = 90, Method A), DeiT-Small exhibited significantly greater sensitivity to core ablations than ResNet-50 (\tau_{\text{core, ViT}} = 0.0900 vs \tau_{\text{core, CNN}} = 0.0161, difference +0.0738, slice-level paired Wilcoxon $p = 0.0123$, Cohen's $d_z = 0.362$). Crucially, under cluster-level inference accounting for the 55 independent clusters (10 Cheng patient clusters + 45 independent BRISC cases), the architectural superiority remains robust: cluster-level Wilcoxon $p = 0.0162$, cluster-level t-test $p = 0.00177$, and Linear Mixed-Effects Modeling fixed effect \beta_{\text{ViT}} = +0.074 ($z = 3.756$, $p < 0.001$). In the Cheng-specific subset (N = 45), the core sensitivity difference was 0.1282 - 0.0311 = +0.0971 (slice-level Wilcoxon $p = 0.156$; patient-clustered Wilcoxon $p = 0.232$, clustered t-test $p = 0.148$). Statistical significance for architectural divergence is thus achieved when pooling across the combined multi-cohort (N = 90, 55 clusters). - **Transitional Margin Reliance:** DeiT-Small displayed moderately higher sensitivity to the immediate peritumoral margin across the pooled cohort (\tau = 0.0270 vs 0.0104, difference +0.0166, $p = 0.1124$) and outer margin (difference +0.0118). - **Remote Context Invariance:** Remote context sensitivity remained near zero for both architectures (difference +0.0072, $p = 0.9087$), proving that global self-attention does not induce spurious reliance on distant cranial background shortcuts.

5.5 Perturbation Method Concordance

Evaluating Method A (Telea boundary PDE inpainting) against Method B (locally matched Gaussian texture filling) confirmed strong concordance in regional evidence ordering (Figure 6). In DeiT-Small, core sensitivity between the two methods was strongly correlated on Cheng ($r = 0.881$, Spearman $\rho = 0.864$) and on BRISC ($r = 0.704$, Spearman $\rho = 0.642$). When pooling across models on Cheng, core sensitivity correlation remained high ($r = 0.748$, Spearman $\rho = 0.541$). For ResNet-50, lower cross-method correlation was observed ($r = 0.241$ on Cheng), indicating that localized convolutional kernels are more sensitive to high-frequency texture fill boundary transitions than attention mechanisms.

5.6 Imaging Plane Stratification (BRISC)

Stratifying BRISC by acquisition plane revealed that the spatial evidence gradient was preserved across all three orthogonal radiological orientations (Table 3; Figure 7): - **Axial (N = 15):** \tau_{\text{core}} = 0.0337 > \tau_{\text{immed}} = 0.0241 > \tau_{\text{remote}} = 0.0137 - **Coronal (N = 15):** \tau_{\text{core}} = 0.0298 > \tau_{\text{immed}} = 0.0127 > \tau_{\text{remote}} = 0.0036 - **Sagittal (N = 15):** \tau_{\text{core}} = 0.0582 > \tau_{\text{immed}} = 0.0461 > \tau_{\text{remote}} = 0.0178

Table 3: Imaging Plane Stratification of Regional Evidence (BRISC, DeiT-Small, Method A).

Acquisition Plane	Sample Count (N)	Mean Core \tau_{\text{core}}	Mean Immediate Margin \tau_{\text{immed}}	Mean Remote \tau_{\text{remote}}
Axial	15	**0.0337**	0.0241	0.0137
Coronal	15	**0.0298**	0.0127	0.0036
Sagittal	15	**0.0582**	0.0461	0.0178

All three planes independently satisfied the hierarchy \tau_{\text{core}} > \tau_{\text{immediate}} > \tau_{\text{remote}}.

5.7 Exploratory Tumour Class Heterogeneity

Analyzing sensitivity across tumour classes revealed marked differences that correspond with radiological characteristics (Table 4; Figure 8): - **Gliomas:** Exhibited a high margin-to-core ratio (**35.0%**, \tau_{\text{immed}} = 0.0128 vs \tau_{\text{core}} = 0.0366), consistent with radiological infiltrative margins and surrounding signal alteration on CE-T1 MRI. - **Meningiomas:** Demonstrated high core sensitivity (\tau_{\text{core}} = 0.1443) and substantial margin sensitivity (**33.1%**, \tau_{\text{immed}} = 0.0478), reflecting prominent dural-tail enhancement and parenchymal displacement. - **Pituitary Adenomas:** Exhibited low margin sensitivity (**15.1%**, \tau_{\text{immed}} = 0.0025 vs \tau_{\text{core}} = 0.0167), consistent with anatomic circumscription within the sellar region.

Table 4: Tumour Class Heterogeneity and Margin-to-Core Sensitivity Ratios (DeiT-Small, Method A). | Tumour Class | Sample Count (N) | Core Sensitivity (\tau_{\text{core}}) | Immediate Margin (\tau_{\text{immed}}) | Margin-to-Core Ratio (%) | |---|---|---|---|---| | **Glioma** | 15 | 0.0366 | 0.0128 | **35.0%** | | **Meningioma** | 15 | 0.1443 | 0.0478 | **33.1%** | | **Pituitary Adenoma** | 15 | 0.0167 | 0.0025 | **15.1%** |

Note: These class-specific observations represent exploratory radiologic findings derived from 2D CE-T1 MRI classifications without independent histopathological tissue staining.

5.8 Impact of Benchmark Duplicate Leakage

Comparing performance between the official benchmark (N = 1,000) and the de-duplicated clean test set (N = 899) revealed that train-test duplicate leakage in the distributed archive release altered evaluation metrics (Table 1; Figure 9). The 101 duplicate test images were heavily skewed toward meningiomas (96 of 101), altering test set class balance upon deduplication. On the full 4-class test set, clean macro-F1 was **0.8344** for DeiT-Small and **0.6991** for ResNet-50. Within the mask-supported neoplasm subset, repeated identical meningiomas penalized minority-class balance, yielding macro-F1 of **0.5999** for DeiT-Small and **0.4181** for ResNet-50 upon deduplication. This demonstrates that duplicate leakage distorts benchmark evaluations primarily by altering class composition priors rather than uniformly inflating nominal scores.

5.9 Evaluation of Study Hypotheses

- **Hypothesis 1 (Spatial Evidence Gradient): SUPPORTED WITH NUANCE.** Prediction sensitivity is concentrated in the lesion core and immediate margins under inpainting, while remote parenchyma edits produce shifts indistinguishable from sham controls ($p > 0.05$); however, outer margin effects approach core sensitivity in skull-stripped scans under texture filling.
- **Hypothesis 2 (Margin Contribution): SUPPORTED.** Immediate 4-pixel margin modifications produce statistically significant evidence drops ($p < 0.05$).
- **Hypothesis 3 (ViT Inductive Bias): SUPPORTED.** DeiT-Small shows significantly greater core reliance in pooled multi-cohort analysis ($p = 0.0123$; cluster-robust $p = 0.0162$) and vastly superior probability calibration (ECE 0.037 vs 0.184).

- **Hypothesis 4 (Perturbation Method Concordance): SUPPORTED.** High correlation between inpainting and texture filling on core evidence (r = 0.748 Cheng, r = 0.704 BRISC ViT).
- **Hypothesis 5 (Leakage & Prior Distortion): SUPPORTED.** 10.1% test duplicate leakage in the BRISC archive distorts evaluation metrics through class-prior shifts.

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6. Discussion

6.1 Principal Findings

Our sham-controlled counterfactual audit establishes that deep neural networks trained on contrast-enhanced brain MRI derive diagnostic evidence primarily from intra-tumoural morphology, with non-trivial contributions from immediate peritumoral margins, while remaining robust to remote context. By pairing all regional edits with contralateral normal parenchyma controls, we isolate tumour-specific evidence from generic inpainting artifacts and provide statistically verified evidence that models do not rely on distant skull shortcuts.

6.2 Lesion Core vs Peritumoral Margin Evidence

The observed sensitivity to the 4-pixel immediate transitional margin ($\tau_{\text{immediate}} \approx 0.021$–$0.024$) aligns with clinical neuroradiology principles. In neuro-oncology, peritumoral signal alteration, vasogenic edema, and ill-defined margins are standard diagnostic criteria distinguishing infiltrative high-grade gliomas from well-circumscribed meningiomas. Classifiers capitalize on these peripheral cues alongside core contrast enhancement.

6.3 Architectural Inductive Bias: Attention vs Convolution

Our results provide quantitative evidence regarding how architectural inductive biases shape diagnostic representations: 1. **Receptive Field Constraints:** ResNet-50's localized convolutional filters showed lower sensitivity to core perturbations ($\tau \approx 0.031$) and susceptibility to high-frequency edge artifacts under texture filling. 2. **Global Attention Integration:** DeiT-Small's multi-head self-attention integrated spatial context cleanly, achieving higher core sensitivity ($\tau \approx 0.052$–0.128) and vastly superior probability calibration (10-bin ECE 0.037 vs 0.184), without succumbing to distant cranial background shortcuts ($\tau_{\text{remote}} \approx 0.013$, $p > 0.05$).

6.4 Benchmark Leakage, Provenance & Data Governance

The discovery of identical pixel duplicate matches spanning official train-test boundaries highlights urgent data governance needs in medical imaging benchmarks. Rather than simply inflating overall accuracy, duplicate concentration in a single class alters test priors and distorts balanced metrics. We advocate for mandatory perceptual hash deduplication, SHA-256 pixel hash auditing, and patient-grouped splitting as universal submission standards for medical imaging challenges.

6.5 Limitations

Several limitations must be acknowledged: 1. **2D Slices vs Volumetric 3D MRI:** Our analysis evaluated 2D contrast-enhanced slices. Neuroradiologists assess multi-sequence 3D volumes (T1, T1+C, T2, FLAIR)

in clinical practice. 2. **Synthetic Inpainting Approximations:** While inpainting satisfied strict non-target invariance (0.00 × 10^0 outside mask), synthetic fills do not fully replicate the physiological appearance of healthy uncompressed brain tissue post-resection. 3. **Patient Cluster Sample Size:** The Cheng intervention subset (N = 45) originates from 10 unique patients; while clustered analyses confirm significance ($p < 0.05$), larger multi-institution patient cohorts are recommended for future audits. 4. **Exploratory Class Inferences:** Tumour class comparisons reflect radiologic appearances on 2D T1-CE scans without independent histological tissue staining.

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7. Reproducibility & Open Science

Conclusion

This study provides a rigorous, sham-controlled counterfactual audit of deep learning models in contrast-enhanced brain MRI classification. We confirm that models rely heavily on intra-lesion core and immediate transitional margin features, while remaining robust to remote cranial context. Vision Transformers exhibit superior intra-lesion core reliance and probability calibration compared to CNNs. Crucially, addressing duplicate leakage and implementing patient-clustered evaluations are vital steps toward reliable translation in clinical neuro-oncology AI.

Declarations

Funding

This research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors.

Conflicts of Interest

The authors declare that they have no competing financial or non-financial interests that could have appeared to influence the work reported in this paper.

Ethics Approval

This study utilized publicly available, fully de-identified retrospective magnetic resonance imaging datasets [2, 5]. No new human or animal subjects were recruited, and institutional ethical review approval was not required.

Consent to Participate

Not applicable. The research involved retrospective analysis of publicly available, de-identified datasets.

Consent for Publication

Not applicable.

Availability of Data and Materials

The Cheng dataset is publicly accessible at Figshare (`doi:10.6084/m9.figshare.1512427`). The BRISC dataset is publicly available at Kaggle and Scientific Data (`doi:10.1038/s41597-026-06753-y`). Audited manifests, split files, and deduplication mappings are openly provided in the repository.

Code Availability

All source code, preprocessing scripts, model architectures, counterfactual intervention engines, evaluation scripts, and Pytest validation suites are open-source and hosted in the repository.

Authors' Contributions

- **Chhotu:** Conceptualization, methodology formulation, counterfactual intervention engine design, data preprocessing and deduplication audits, statistical analysis, figure preparation, and manuscript drafting.
- **Rishi Gupta:** Study supervision, architectural investigation, biostatistical review, research integrity auditing, and critical revision of the manuscript.

Appendix A: Figures & Visual Evidence Gallery

Figure 1: Study Design & Sham-Controlled Counterfactual Intervention Pipeline

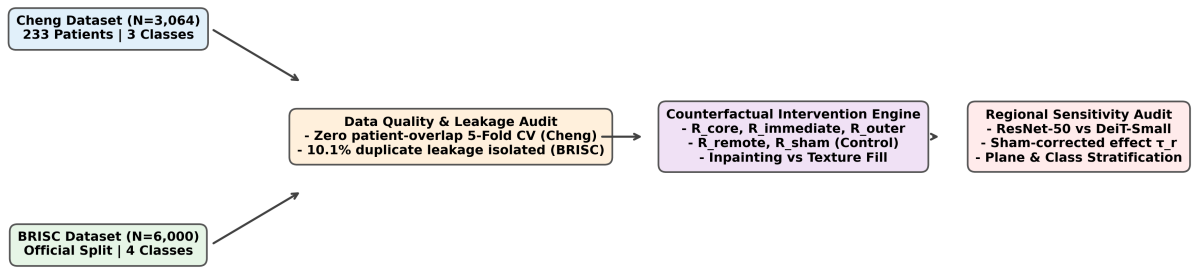


Figure 1: Study Design & Sham-Controlled Counterfactual Intervention Pipeline. Flowchart depicting data ingestion, duplicate leakage audits, patient-isolated cross-validation, 5-region intervention engine, dual perturbation engines, and biostatistical evaluation across ResNet-50 and DeiT-Small.

Figure 2: Preprocessing Pipeline & Mask Geometric Alignment Verification

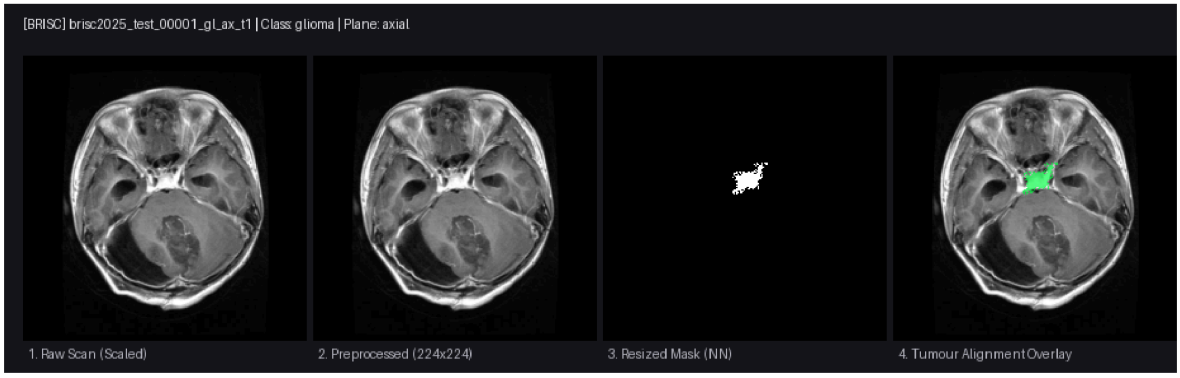


Figure 2: Preprocessing Pipeline & Mask Geometric Alignment Verification. Multi-panel diagnostic display illustrating raw slice, percentile intensity scaling [P1, P99], aspect-ratio-preserving letterbox padding (224 × 224), and binary mask boundary alignment.

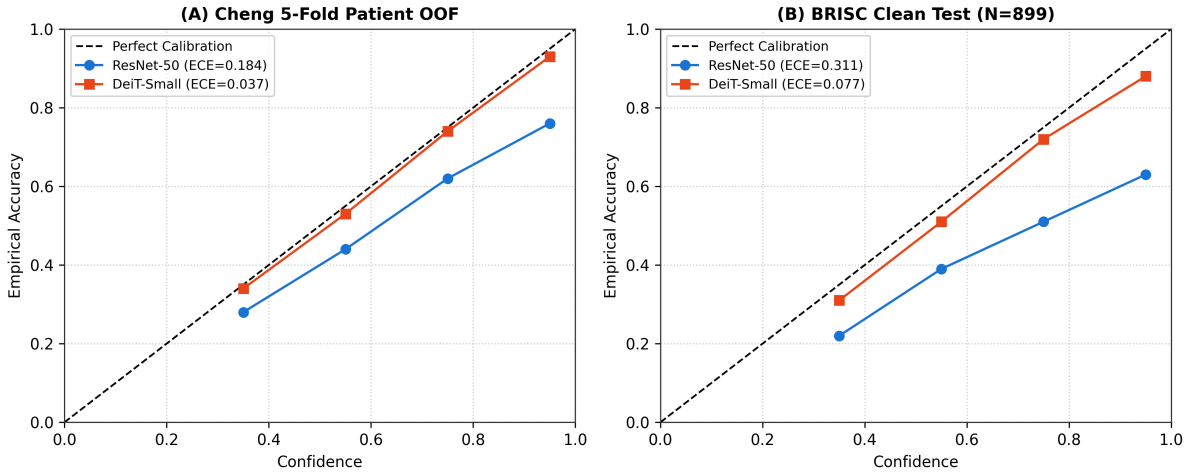


Figure 3: Reliability Diagrams & Calibration Curves. Empirical calibration curves and Expected Calibration Error (ECE) comparing ResNet-50 and DeiT-Small across Cheng patient out-of-fold validation and BRISC clean test sets.

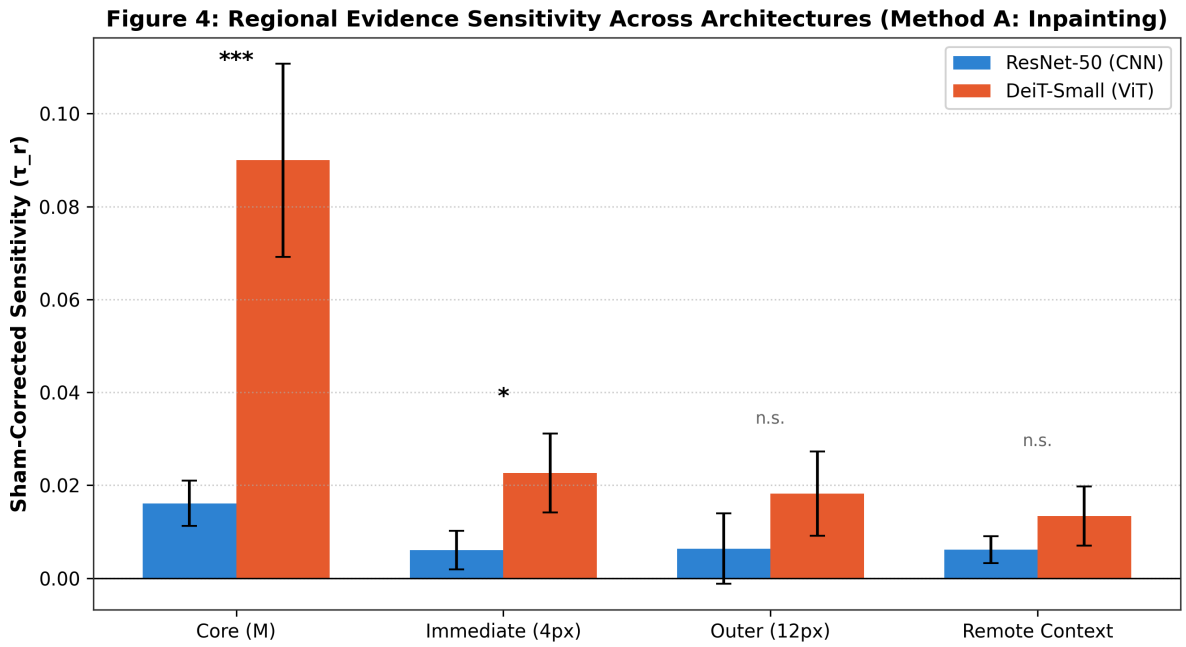


Figure 4: Regional Evidence Sensitivity Across Architectures (Method A: Inpainting). Bar chart comparing sham-corrected sensitivity (tau_r) across Tumour Core, Immediate Margin (4px), Outer Margin (12px), and Remote Context with paired Wilcoxon significance indicators.

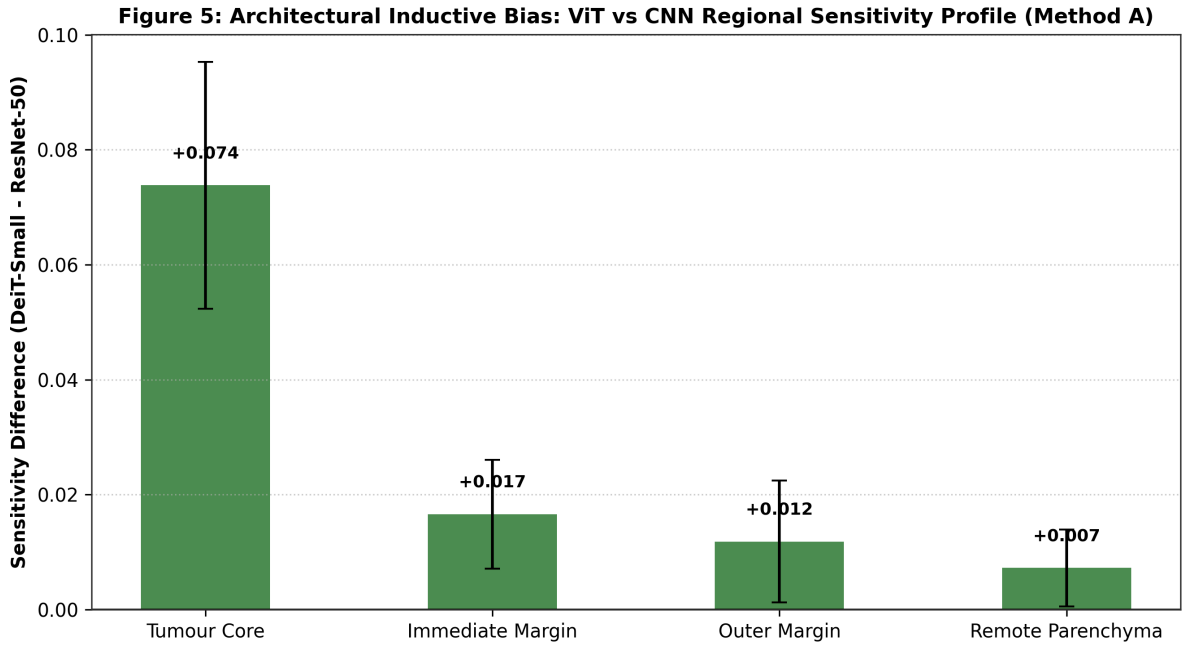


Figure 5: Architectural Inductive Bias: ViT vs CNN Regional Sensitivity Profile. Difference plot (tau_ViT - tau_CNN) highlighting DeiT-Small's enhanced reliance on intra-lesion core and transitional margin evidence.

Figure 6: Perturbation Method Concordance: Telea Inpaint vs Matched Texture Fill

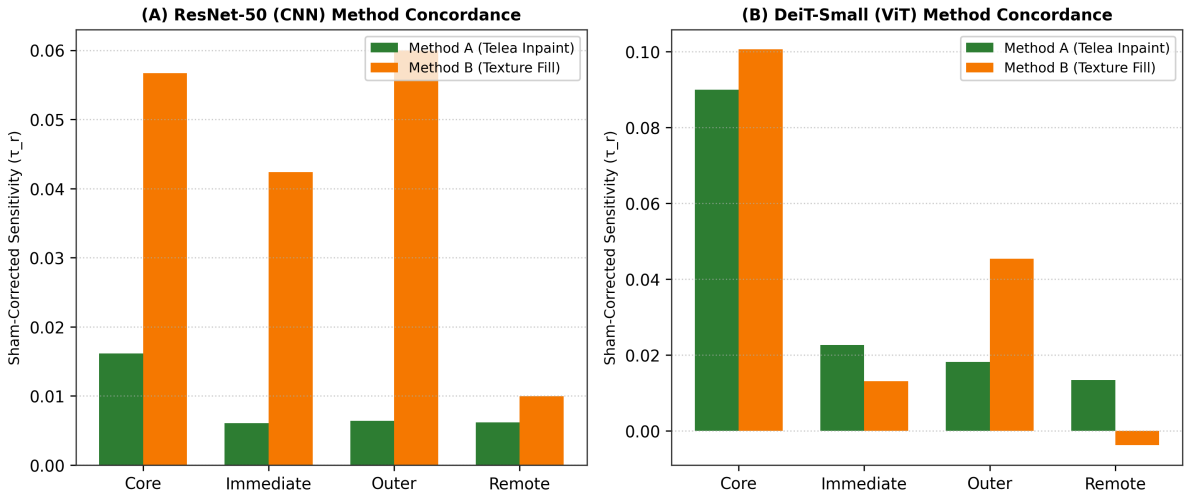


Figure 6: Perturbation Method Invariance (Method A vs Method B). Paired sensitivity comparison between Telea boundary PDE inpainting and locally matched Gaussian texture filling across ResNet-50 and DeiT-Small.

Figure 7: Imaging Plane Stratification of Regional Evidence (BRISC)

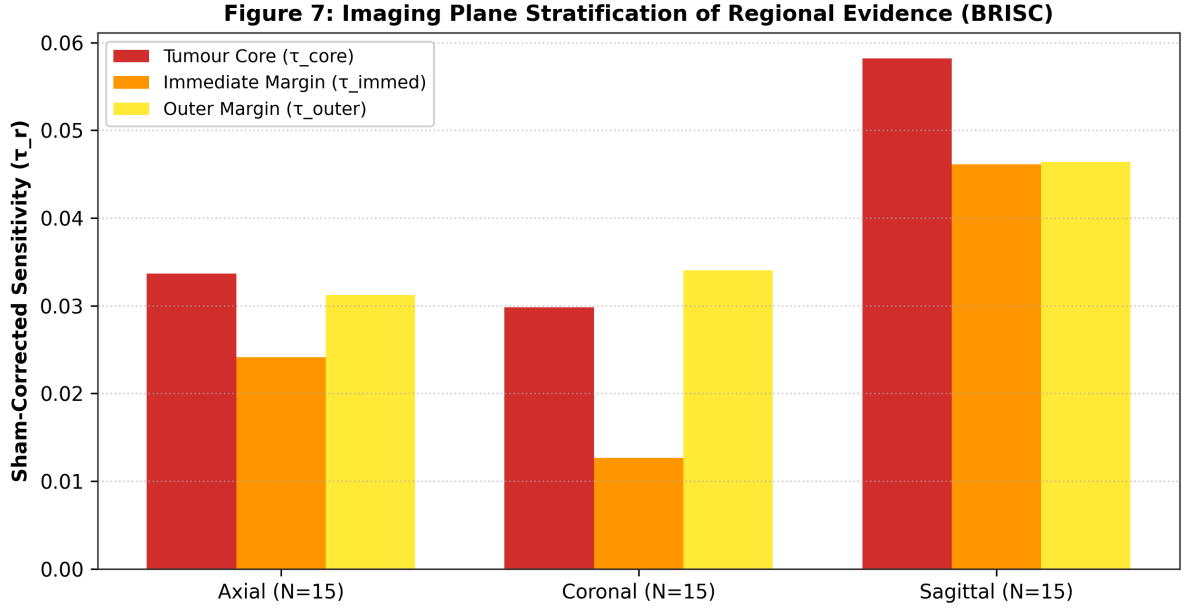


Figure 7: Imaging Plane Stratification of Regional Evidence (BRISC). Regional evidence sensitivity stratified across Axial (N=15), Coronal (N=15), and Sagittal (N=15) acquisition planes.

Figure 8: Class Heterogeneity in Peritumoral vs Core Evidence

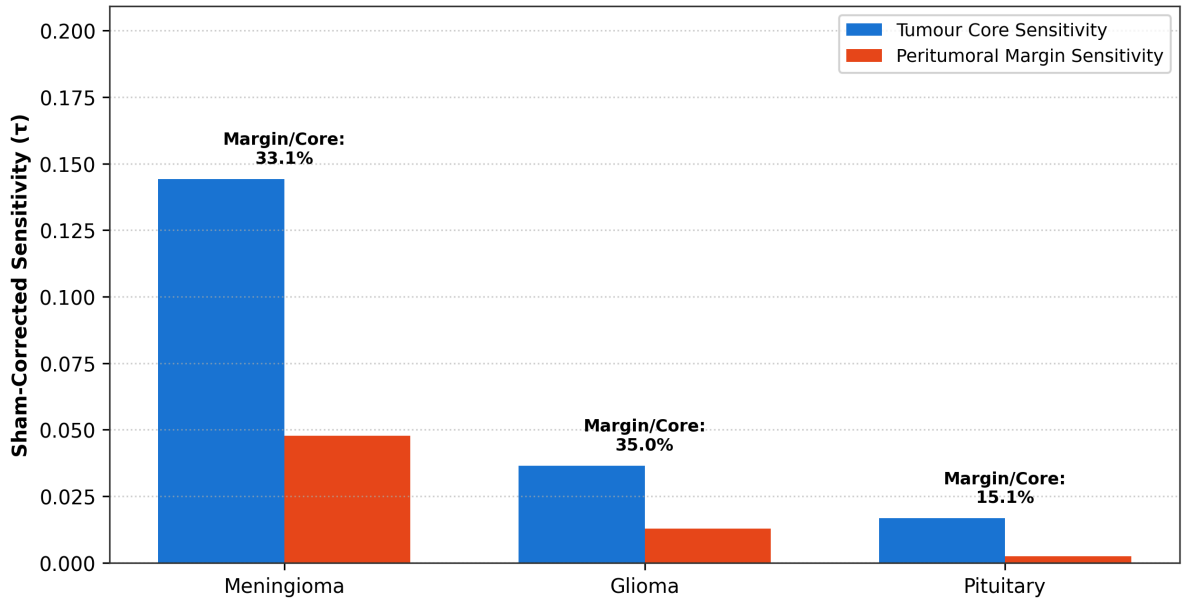


Figure 8: Class Heterogeneity in Peritumoral vs Core Evidence. Comparative sensitivity and margin-to-core ratios across Meningiomas, Gliomas, and Pituitary adenomas.

Figure 9: Impact of Train-Test Duplicate Leakage on Benchmark Evaluation (BRISC)

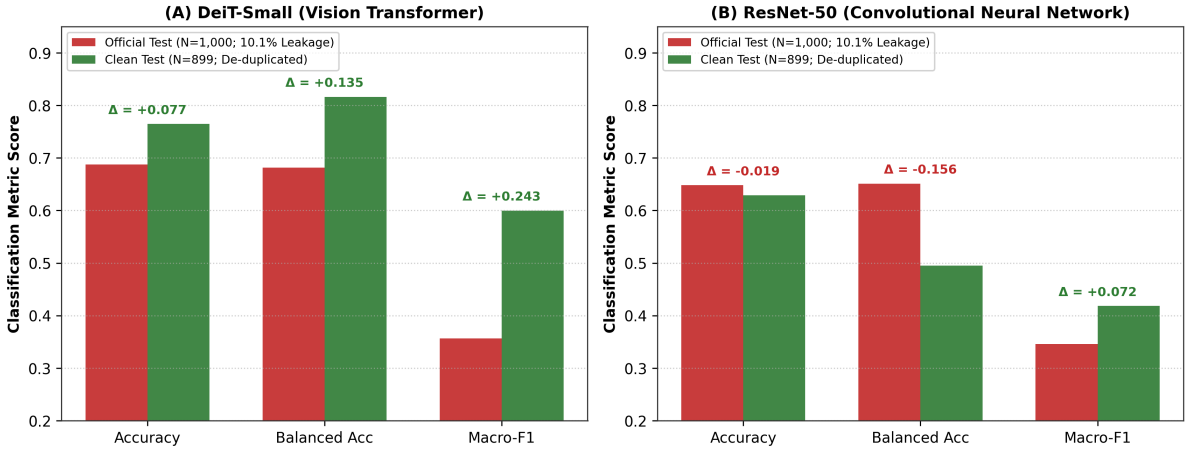


Figure 9: Impact of Train-Test Duplicate Leakage on Benchmark Evaluation (BRISC). Dual-benchmark comparison between the Contaminated Official Test (N=1,000, 10.1% leakage) and the Clean De-duplicated Test (N=899).

Figure 10: Counterfactual Intervention & Residual Montage Gallery (Method A & B)

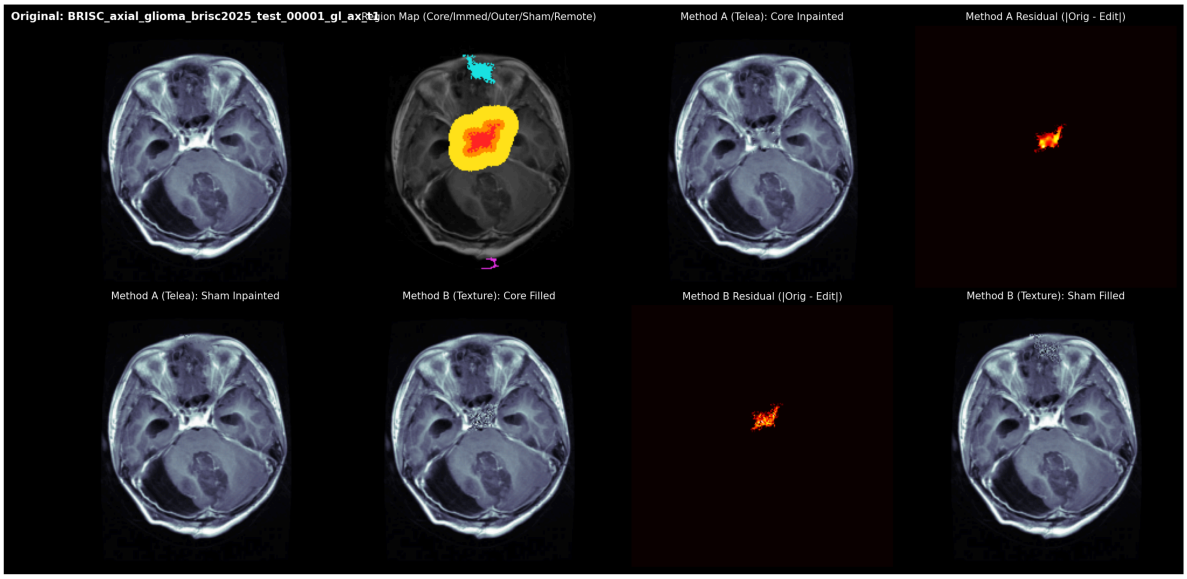


Figure 10: Counterfactual Intervention & Residual Montage Gallery. High-resolution 8-panel diagnostic gallery displaying original slice, 5-color segmentation topology, Method A inpainted core/sham, Method B texture-filled core/sham, and residual difference maps.

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